

Name - Nirupom Saha
Reg. No - 2141019011

Linux System Programming

Task 1: Kernel Architecture diagram with short description of each major component

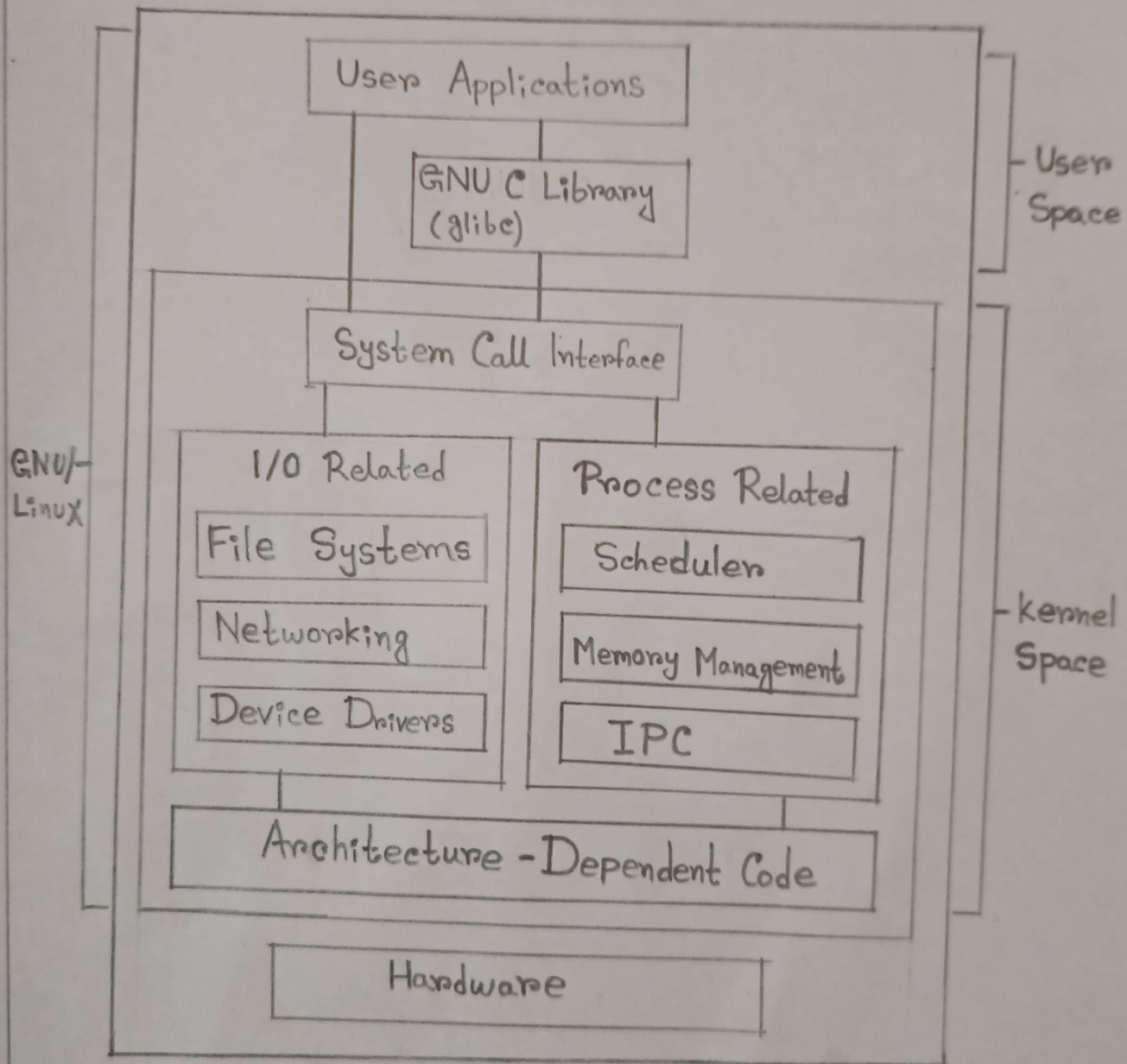


Fig: Linux kernel Architecture

Scheduler: The scheduler is responsible for managing process execution by deciding which process runs on the CPU and for how long. It ensures fair CPU time distribution among processes based on their priorities.

Memory Management: This component handles both physical and virtual memory. It manages memory allocation, paging, swapping, and ensures memory protection and efficient usage across processes.

File System: The file system provides a way to store, retrieve, and manage data on storage devices. It abstracts storage hardware and supports multiple file system formats like ext4, XFS, and FAT.

Network Stack: The network stack implements networking protocols and manages communication over networks. It supports layered protocols like TCP/IP, allowing data exchange between systems.

Architecture-Dependent Code: This section contains code optimized for specific hardware architectures (e.g., x86, ARM). It handles low level tasks such as CPU initialization, interrupt handling, and context switching.

Process Management: Process management oversees process creation, scheduling, and termination. It enables communication and synchronization between processes using mechanisms like signals, pipes, and IPC.

Device Drivers: Device drivers act as intermediaries between the kernel and hardware devices. They provide standard interfaces to access and control hardware like keyboards, disks, and network cards.

System Call Interface: The system call interface provides a controlled entry point for user-space application to access kernel services. It ensures secure and managed communication between user space and the kernel.